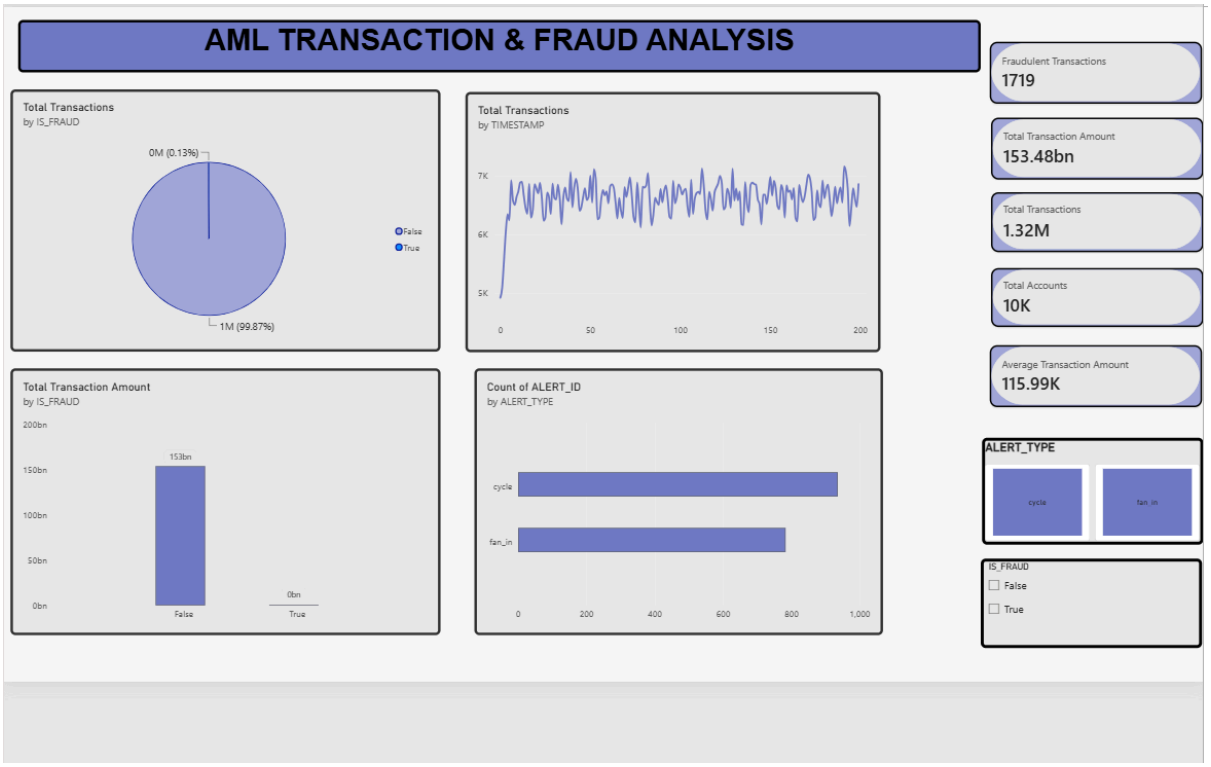
AML Transaction & Fraud Analysis – Business Insight Report



The dashboard presents an overview of Anti-Money Laundering (AML) transactions, fraud status, transaction activity, and alert types. It includes KPI cards, charts, and interactive filters to help users identify suspicious activities and monitor transaction patterns.

2. Overview of the AML Dataset

The AML (Anti-Money Laundering) dataset contains transaction-related information used to analyse financial activities, identify potential fraud, and monitor suspicious transaction patterns. The dashboard uses fields such as IS_FRAUD, TIMESTAMP, ALERT_ID, and ALERT_TYPE.

The main purpose of the dashboard is to monitor transaction volumes, compare fraudulent and non-fraudulent transactions, analyse alert categories, and support financial institutions in identifying activities that may require further investigation.

3. KPI Interpretation

The dashboard displays five key performance indicators (KPIs).

KPI	Dashboard Value	Interpretation

Fraudulent Transactions	1,719	The number of transactions identified as fraudulent in the dataset.
Total Transaction Amount	153.48 billion	The total monetary value of transactions represented in the dashboard.
Total Transactions	1.32 million	The overall volume of transactions recorded.
Total Accounts	10,000	The total number of accounts represented.
Average Transaction Amount	115.99K	The average monetary value per transaction, as displayed in the dashboard.

These KPIs provide a quick overview of transaction activity and potential fraud. The number of fraudulent transactions helps assess the scale of detected fraud, while total transaction value and average transaction amount help evaluate the financial activity represented in the dataset.

Note: The KPI values are based on the supplied screenshot. The underlying calculations and aggregation settings should be verified before using them for formal reporting.

4. Fraud Status Analysis

The pie chart compares transactions according to the IS_FRAUD field.

- False: Approximately 99.87% of transactions are classified as non-fraudulent.
- True: Approximately 0.13% of transactions are classified as fraudulent.

The chart indicates that most transactions are classified as non-fraudulent, while a very small proportion is identified as fraudulent. Although the fraud percentage is low, fraudulent transactions can cause significant financial losses and therefore require careful monitoring.

The bar chart showing total transaction amount by fraud status indicates that the non-fraudulent category accounts for almost the entire displayed transaction value. The fraudulent category appears close to zero at the chart's current scale.

Insight: A low fraud percentage does not necessarily mean that the risk is insignificant. The financial impact and characteristics of the fraudulent transactions should also be examined.

5. Transaction Activity Analysis

The line chart displays transaction activity by timestamp. After an initial increase, the plotted values fluctuate around approximately 6,000–7,000.

This pattern suggests that transaction activity remains relatively consistent over much of the displayed period, with regular fluctuations. The initial increase may reflect a change in activity at the beginning of the period.

Monitoring transaction activity over time can help identify unusual spikes, sudden drops, or changes in transaction behaviour that may require further investigation.

Insight: A time-based analysis can help AML teams identify unusual transaction patterns and investigate periods with abnormal activity.

6. Alert Analysis

The horizontal bar chart compares the number of alerts across two alert types: cycle and fan_in.

- Cycle alerts: Approximately 930 alerts.
- Fan-in alerts: Approximately 780 alerts.

Cycle alerts have a higher count than fan-in alerts in the displayed dashboard. This indicates that cycle-related patterns are more frequently represented among the alerts shown.

In AML monitoring, cycle patterns may involve funds moving through a sequence of accounts, while fan-in patterns may involve funds from multiple sources converging into a single account. These patterns can have legitimate explanations, so an alert should be investigated rather than automatically treated as proof of fraud.

Insight: Cycle alerts deserve attention because they form the larger category in this visual. Both alert types should be reviewed based on their transaction details, risk level, and supporting evidence.

7. Filter-Based Analysis

The dashboard includes two interactive filters that allow users to explore the data.

Alert Type filter

The ALERT_TYPE filter allows users to select categories such as cycle and fan_in. Selecting an alert type helps compare the relevant transactions and investigate the patterns associated with that category.

Fraud Status filter

The IS_FRAUD filter allows users to select False or True. Selecting True helps focus the analysis on transactions classified as fraudulent, while selecting False helps examine non-fraudulent transactions.

These filters make the dashboard more useful by allowing users to explore specific categories instead of relying only on overall totals. The results should be checked to confirm that all relevant charts and KPI cards respond correctly to each filter selection.

8. Key AML Insights

The following insights can be drawn from the dashboard:

1. Low proportion of fraudulent transactions: Only around 0.13% of transactions are classified as fraudulent, compared with approximately 99.87% classified as non-fraudulent.

2. Large transaction volume: The dataset contains approximately 1.32 million transactions, making automated monitoring valuable for reviewing activity at scale.

3. Significant transaction value: The total transaction amount is displayed as 153.48 billion, indicating a substantial volume of financial activity represented in the dashboard.

4. Cycle alerts are more frequent: Cycle alerts outnumber fan-in alerts in the displayed chart and may require focused investigation.

5. Fluctuating transaction activity: The time-series chart shows repeated fluctuations after an initial increase, making continuous monitoring useful.

6. Interactive analysis supports investigation: The filters allow users to examine fraud status and alert types separately, helping narrow down areas for further review.

7. Further verification is necessary: The dashboard provides useful indicators, but the underlying transactions must be reviewed before concluding that suspicious activity represents confirmed money laundering or fraud.

9. Recommendations

Based on the dashboard analysis, the following recommendations are suggested.

- Strengthen fraud monitoring: Use automated monitoring rules to flag unusual transaction amounts, frequencies, and account behaviours.

- Investigate high-priority alerts: Review cycle and fan-in alerts using transaction histories, account relationships, and other relevant evidence.

- Monitor transaction activity continuously: Use timestamp-based analysis to identify unusual increases, repeated patterns, or unexpected changes in activity.

- Improve fraud detection: Review the characteristics of confirmed fraudulent transactions to identify useful patterns for improving detection rules.

- Use risk-based investigation: Prioritise alerts according to factors such as transaction value, frequency, unusual account relationships, and supporting evidence.

- Improve dashboard accuracy: Verify KPI calculations, chart aggregation, and filter interactions. In particular, confirm that the average transaction amount is calculated using the correct transaction count and that the time-series chart uses actual timestamps.

- Maintain proper records: Document alert investigations, decisions, and supporting evidence to improve accountability and compliance.

- Review false positives: Investigate alerts that do not lead to confirmed suspicious activity and refine monitoring rules where appropriate, without weakening necessary controls.

10. Conclusion

The AML Transaction & Fraud Analysis dashboard provides a useful overview of transaction activity, fraud classification, and alert patterns. It indicates that most transactions are classified as non-fraudulent, while a small proportion is flagged as fraudulent. It also shows that cycle alerts are more frequent than fan-in alerts and that transaction activity fluctuates over time.

The dashboard's KPIs, charts, and interactive filters support the monitoring and investigation of potentially suspicious transactions. By improving automated detection, prioritising high-risk alerts, monitoring transaction patterns, and validating dashboard calculations, financial institutions can strengthen their AML monitoring processes.

Overall, the dashboard can support better financial risk management and compliance by helping analysts identify unusual patterns for further investigation. However, dashboard alerts are indicators for investigation, not conclusive proof of money laundering or fraud.